

6.H1 Proving All Circles Are Similar

Lesson Introduction

 Benchmark	MA.912.GR.6.5 Apply transformations to prove that all circles are similar.		 Learning Target	I can justify that two circles are similar and conclude that any circle can be mapped to another circle using similarity transformations.
 Benchmark Coherence	Building On		Working Toward	
	N/A		N/A	
 Essential Questions	- How do rigid and nonrigid transformations prove that all circles have similarity? - Why does a circle follow the same rules as other shapes regarding rigid transformation?			
 Lesson Narrative	This lesson explores rigid and nonrigid transformations of circles. Students explore how one circle can be mapped onto other circles that are either congruent or the result of an enlargement or reduction. Conclusions are drawn that all circles are similar but not necessarily congruent.			
 Component Summary	6.H1.1 Warm-Up (~5 min.)	Notice and Wonder about overlapping circles (<i>SE p. 318</i>)	6.H1.3 Your Turn! (10-15 min.)	Transforming a circle to create a similar circle (<i>SE p. 320</i>)
	6.H1.2 Exploration (15-20 min.)	Mapping a circle onto three other circles to determine similarity (<i>SE p. 318</i>)	6.H1.4 Wrap-Up (~5 min.)	Reflecting on the interest and understanding of the lesson (<i>SE p. 321</i>)
 Resources	Glossary	Materials		Additional Preparation
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Similar - Congruent	(Optional) Masking Tape (Optional) Circular Objects		- This is an honors lesson. Students in regular Geometry do not need to complete this lesson. - For 6.H1.2 Exploration or 6.H1.3 Your Turn!, if you are considering using the classroom floor as a grid, set this up prior to class. Consider using a ruler to place sticky notes one foot from a wall, then roll the tape out, and repeat this process measuring one foot from the last piece of tape.

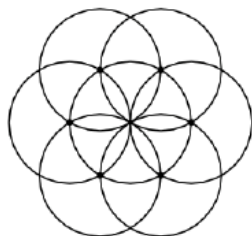
 Teacher Tips	Common Misconceptions	Things to Consider
	Students may think that congruent circles are not similar.	Consider creating a large grid on the floor and using yarn, hula-hoops, or other round objects to recreate the circles in 6.H1.2 Exploration or 6.H1.3 Your Turn! Then, students could physically move the circles as suggested.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time For 6.H1.3 Your Turn! question 1b, consider implementing this routine. This structure provides a purpose for student conversations as well as fortifies student output. Throughout the process, students should be pressed to create a stronger (better justifications and examples) and clearer (with more precise terms and linked, organized, complete sentences) response.	MA.K12.MTR.5.1: Patterns and Structure This lesson is designed to guide students through relating previously learned concepts to the new concept of circles. From 6.H1.2 Exploration, students are using the same vocabulary of mapping circles as they did when mapping polygons earlier in Unit 6.
 Differentiate Instruction	Students will be able to activate background knowledge at 6.H1.2 Exploration. This initially occurs when students are asked to “map” one circle to another. This is the same terminology as earlier in the unit, and the cadence of this lesson mimics previous lessons, when students learned to map and identify transformations.	
Lesson Notes:		

6.H1.1 Warm-Up: Notice and Wonder

Component Overview: Students notice and wonder about overlapping circles and how they could create unique and artistic patterns and shapes.



6.H1.1 Warm-Up: Notice and Wonder



Consider asking the following probing questions:

- If we remove or color in parts of this image, what naturally occurring shapes may you find? Do you see a flower? Do you see a sand dollar? Do you see a star fish?

1. What do you notice?

Answers will vary. I see 7 overlapping circles.

2. What do you wonder?

Answers will vary. I wonder how somebody thought to arrange circles in this artistic way.

6.H1.2 Exploration: Are Circles Similar?

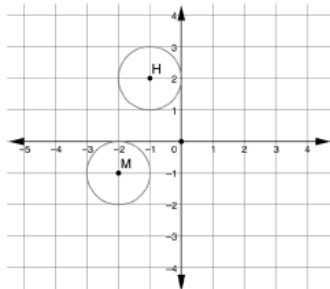
Component Overview: Students map a given circle onto three different circles using rigid and nonrigid transformations to determine that all circles are similar but not necessarily congruent.



6.H1.2 Exploration: Are Circles Similar?

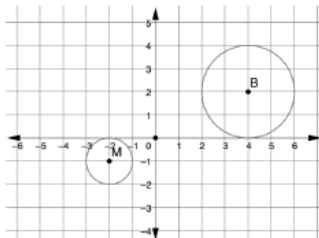
Answer the following questions with your partner.

1. Circle M and Circle H are shown on the coordinate plane.



- a. How can Circle M be mapped onto Circle H ?
Answers will vary. Translate Circle M right 1 unit and up 3 units.
- b. What can you conclude about Circle M and Circle H ?
Answers will vary. All points on Circle M map to all points on Circle H . Therefore, the circles are congruent.

2. Circle M and Circle B are shown on the coordinate plane.



- a. How can Circle M be mapped onto Circle B ?
Answers will vary. Circle M can be mapped onto Circle B by translating it right 6 units and up 3 units, then dilating the result by a scale factor of 2 centered at $(4, 2)$.



Consider asking the following while monitoring student work on question 1.

- What is another way to verify that two circles are congruent? (Answer: they have the same radius.)
- Are there other ways to map Circle M onto Circle H ? (Possible answer other than the listed answer: students may also rotate Circle M 90° clockwise about the origin.)



Consider asking the following while monitoring student work on question 2.

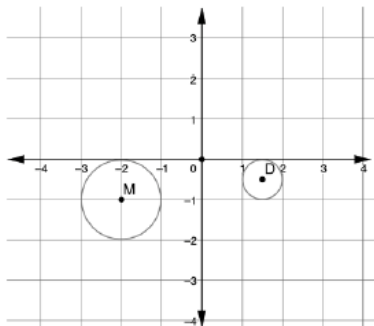
- How do I know that the two circles are not congruent? (Answer: they don't have the same radius.)
- Are there other ways to map Circle M onto Circle B ? (Possible answer other than the listed answer: students may also rotate Circle M 180° about the origin and then dilate by 2 about the origin.)

6.H1.2 Exploration: Are Circles Similar? (continued)

- b. What can you conclude about Circle M and Circle B ?

The circles are similar but not congruent.

3. Circle M and Circle D are shown on the coordinate plane.



- a. How can Circle M be mapped onto Circle D ?

Answers will vary. Circle M is translated right 3.5 units, up 0.5 unit. The result is then dilated using a scale factor of 0.5 with Point D as the center of dilation.

- b. What can you conclude about Circle M and Circle D ?

They are similar, but not congruent.

4. What can you conclude about Circle M , Circle H , Circle B , and Circle D ?

Because Circle M can be mapped to all three circles, we can conclude that they are all similar.

5. Complete the statement.

- ☐ All
☐ Some
☐ None

circles are

- ☐ congruent
☒ similar

because any

circle can be mapped to another circle using similarity transformations.



Consider asking the following while monitoring student work on question 5.

- What transformations are considered “congruence” transformations? (Translations, reflection, rotation.)
- What transformations are considered “similarity” transformations? (Translation, reflection, rotation, and dilation.)

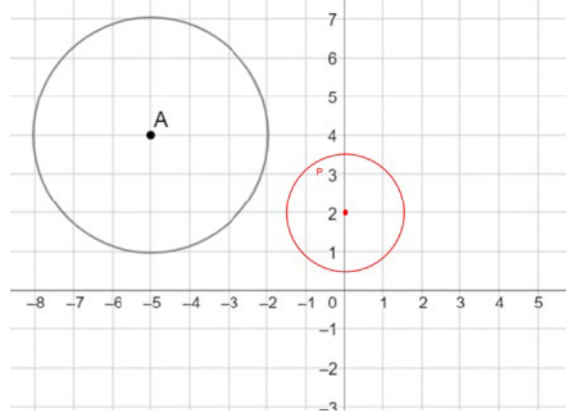
6.H1.3 Your Turn!

Component Overview: Students draw a similar circle to a given circle using rigid and nonrigid transformations and explain why they are similar using the word “transformation.”



6.H1.3 Your Turn!

1. Circle *A* is shown on the coordinate plane.



- Draw Circle *P*, the result of a translation 5 units right and 2 units down and a dilation of $\frac{1}{2}$ about the center of Circle *P*.
 - Explain how Circle *A* and Circle *P* are similar. Include the word “transformation” in your explanation.
Circle A and Circle P are similar because a dilation is a transformation that maintains proportionality.
2. Lana and Henry are writing notes from today’s lesson.

Lana’s Notes	Henry’s Notes
All circles are similar; therefore, all circles are also congruent.	All circles are similar, but not all circles are congruent.

Whose notes do you agree with? Why?

Answers will vary. I agree with Henry because, while all circles are similar, not all circles have the same radii and therefore are not congruent.



For a kinesthetic entry-point, consider creating a large coordinate grid on the floor and using yarn, Hula-Hoops, or other circular objects to recreate the circles in this component and have students physically move the circles as suggested.



Stronger and Clearer Each Time

For question 1b, consider implementing this routine. After students have written a response to the question, pair them with another student to refine and clarify their response through conversation. Then, pair students differently and repeat. After several pairings, all students should have a correctly revised response.

6.H1.4 Wrap-Up: Reflection

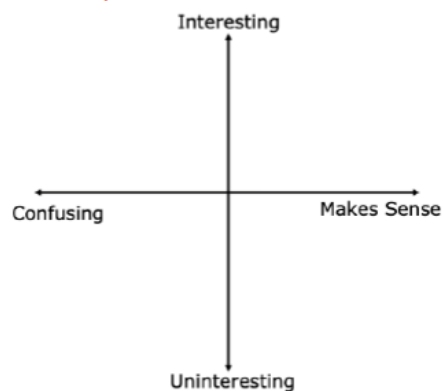
Component Overview: Students plot a point on the grid that best represents their interest and confidence on the lesson.



6.H1.4 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and understanding for today's learning target.

Answers will vary.



Explain the location of your point.

Answers will vary.



Consider using the point a student placed the grid as a jumping-off point for individual conferences about their progress and understanding.